Journal of Health Ethics

Volume 21 | Issue 1 Article 6

2025

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Recommended Citation
Rizvi, M. Z. (2025). Reproductive Autonomy In Immunological Intervention: Ethical Boundaries in Maternal-Fetal Immune Therapy. *Journal of Health Ethics, 21*(1). http://dx.doi.org/10.18785/jhe.2101.06

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# Reproductive Autonomy in Immunological Intervention: Ethical Boundaries in Maternal-Fetal Immune Therapy

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## Abstract
Recent advances in immunology have broken barriers in reproductive medicine through immune-modulating therapies that support fertility and pregnancy. Treatments such as intravenous immunoglobulin (IVIG) and intralipid infusions are increasingly offered to patients with Recurrent Implantation Failure (RIF) or Recurrent Pregnancy Loss (RPL), conditions that result in miscarriages and failed IVF treatments. Maternal-fetal immunology reveals a delicate interplay between immune tolerance and defense: the maternal system must accept a genetically foreign fetus while preserving immunological protection. Disruptions in this balance are implicated in implantation disorders and pregnancy failure. As a result, immune-based therapies have emerged as promising interventions, though they raise ethical concerns about informed consent, medical paternalism, patient safety, and the tension between autonomy and provider responsibility. Despite their potential, patient decision-making may be undermined by misinformation and the emotional vulnerability of grieving parents. Consequently, reproductive immunotherapy occupies an ethically fraught space demanding clearer regulatory oversight and attention to moral considerations. Upholding reproductive autonomy requires more than offering treatment; it necessitates transparent, evidence-based, and ethically sound practices. This paper examines ethical and informed consent challenges in reproductive immunology, focusing on gaps between patient understanding and therapeutic risks, conflicts between maternal and fetal interests, and the commercialization of these therapies in private fertility clinics. It draws on bioethical principles and alternative frameworks such as relational autonomy and the ethics of care.

**Keywords:** Reproductive Autonomy; Maternal-Fetal Immune Therapy; Immune Tolerance; Intravenous Immunoglobulin (IVIG); Intralipid Therapy; Recurrent Pregnancy Loss (RPL); Recurrent Implantation Failure (RIF); Informed Consent; Therapeutic Misconception; Maternal-Fetal Interface

## Introduction
Reproductive success in women is a complex and regulated balance of hormonal, genetic, and immunological factors. While reproductive disorders such as infertility and pregnancy loss have long been attributed to chromosomal abnormalities, endocrine imbalances, or structural defects in the reproductive tract, recent research has revealed a deeper complexity; one in which the immune system plays a vital, though often overlooked, role (Garmendia et al., 2025). This growing body of evidence highlights the immune system not just as a defense against infection but as an active participant in regulating the success or failure of pregnancy. Recurrent pregnancy loss is a condition by which 2 or more consecutive pregnancies are terminated before 24 weeks of gestation, whereas recurrent implantation failure is a condition when 3 or more embryo transfers or 2 egg donations result in unsuccessful pregnancies (Garmendia et al., 2025). For many women facing recurrent implantation failure (RIF) or recurrent pregnancy loss (RPL), the underlying cause may not be anatomical or chromosomal but immunological. In these cases, conventional diagnostic testing often fails to yield answers. The maternal immune system, which is posed with the challenge of protecting the mother from infection while tolerating the semi-allogenic [50% genetics from the maternal genome] fetus, may be the very factor

contributing to reproductive failure. This immunological paradox requires the mother's immune cells to suppress the normal defensive response toward paternal antigens while still functioning in surveillance and repair roles throughout pregnancy. When this finely tuned immune balance is disrupted, the result may be embryo rejection, implantation failure, or improper placental development.

A growing number of studies have implicated specific immune cell subsets, particularly T-helper (Th) cells, regulatory T cells (Tregs), and natural killer (NK) cells, in determining pregnancy outcomes (Garmendia et al., 2025). The overactivation or lack of these cells can lead to an environment of chronic inflammation and rejection at the maternal-fetal interface (Garmendia et al., 2025). The clinical consequences of these immune abnormalities are far-reaching. An inflammatory or cytotoxic immune environment may lead to embryo rejection, hinder implantation, or disrupt placentation [the process by which the placenta attaches to and invades the uterine wall] (Garmendia et al., 2025). These complications may occur even when embryos and uterine anatomy are normal, further highlighting the role of immune dysfunction in unexplained reproductive failure.

As a result, clinicians and researchers have begun to explore immunomodulatory therapies to restore immune balance and support successful pregnancy outcomes in patients with suspected immune-related RIF and RPL. Two of the most commonly used interventions are intravenous immunoglobulin (IVIG) and intralipid therapy, also referred to as intralipid emulsion therapy (ILET). IVIG involves the infusion of concentrated antibodies collected from healthy donors and is thought to promote tolerance by increasing Treg populations, reducing NK cell cytotoxicity, and downregulating pro-inflammatory cytokines. There has been a significant increase in clinical pregnancy rates of about 95% in observational studies for patients with RIF (Satio, 2024). Intralipid therapy, by contrast, uses a fat emulsion to lessen cytotoxic activity of NK cells, and inflammatory responses within the uterus (Satio, 2024). Both treatments have been offered in fertility clinics as adjunctive therapies for women who have exhausted traditional reproductive options (Satio, 2024).

However, the use of these therapies remains controversial, as there is no 'real' treatment for RIF or RPL. While some studies suggest improved implantation and live birth rates among women treated with IVIG or intralipids, particularly those with elevated NK cell activity or abnormal immune profiles, larger randomized trials and meta-analyses have produced inconsistent findings, and standardized diagnostic criteria for initiating such treatments are still lacking (Satio, 2024). Further research is needed to elucidate the precise immune mechanisms involved in RPL and RIF, to develop reliable biomarkers for diagnosis, and to support the development of personalized treatment strategies that address each patient's unique immune landscape. Understanding how the maternal immune system interacts with the developing embryo is not only crucial for explaining cases of unexplained infertility but is also central to the ethical and clinical debate over the growing use of immune-based therapies in reproductive medicine. While intralipid therapy (ILET) and intravenous immunoglobulin (IVIG) are increasingly offered as adjunctive treatments for women experiencing recurrent implantation failure (RIF) or recurrent pregnancy loss (RPL), their use remains largely experimental and off-label in the context of reproductive medicine. This introduces several ethical challenges, chief among them, the potential for therapeutic misconception. This occurs when patients mistakenly interpret a treatment as being definitively effective or curative, even when its efficacy remains unproven (Horng & Grady, 2003). In emotionally vulnerable contexts such as infertility treatment, this misconception can seriously undermine informed consent.

Women desperate to conceive may consent to costly and physically invasive treatments like IVIG or intralipids believing these are standard-of-care interventions, when in reality, the scientific evidence is still evolving and clinical outcomes remain inconsistent. Many fertility clinics offer these therapies in private, often unregulated settings without adequately disclosing their experimental nature, off-label status, or the potential risks (such as allergic reactions, immune suppression, or possible interference with the maternal-fetal immune balance) (Shakeel et al., 2022). For example, a woman undergoing her third failed IVF cycle might be told that intralipid infusions can "calm her immune system" and support

implantation. She may agree to treatment and pay out-of-pocket, unaware that intralipid is not FDA-approved for this use and that large-scale randomized trials have yet to confirm its benefit. Her consent, though seemingly voluntary, may stem more from desperation than informed understanding, thereby failing to meet ethical standards for autonomy and transparency.

Several core bioethical principles are at stake. Autonomy requires that patients make decisions with access to full, transparent, and unbiased information; if treatments are marketed or implied to be more effective than they truly are, that autonomy is gone (Hardman & Hutchison, 2023). Non-maleficence, or the principle of "do no harm," may be compromised when physicians offer interventions with uncertain benefits and possible risks without proper oversight. From a perspective of justice, access to these treatments often depends on financial resources (most are not covered by insurance) raising concerns about health equity (Holm, 1995). Finally, the principle of beneficence demands that care decisions be grounded in science, not emotional urgency or commercial incentives (Holm, 1995). While the intent behind offering these therapies is often to help, when the treatments are insufficiently studied or prematurely commercialized, their use can blur the line between innovation and exploitation.

However, an important counterargument emerges in defense of these therapies, one rooted in a different interpretation of reproductive autonomy. While concerns about therapeutic misconception are valid, some argue that restricting or over-regulating access to experimental immunotherapies risks medical paternalism, where clinicians or institutions dictate what is acceptable care, even when informed and competent patients are willing to take those risks (Chin, 2002). For many women who have exhausted conventional treatments and are rapidly approaching the end of their reproductive window, IVIG or intralipid therapy may represent not false hope, but a reasonable and autonomous decision.

From this perspective, reproductive freedom includes the right to access emerging or investigational therapies, especially in the absence of viable alternatives. If clinicians clearly communicate the limits of current research, and if patients fully understand the potential benefits and risks, then offering such therapies may be not only ethical but empowering. In fact, many women report that being offered these options, even when framed as experimental, provides a sense of agency and emotional relief in the midst of a deeply personal and uncertain process (Chin, 2002). In this view, ethical care involves not just protecting patients from risk, but also respecting their capacity to navigate uncertainty in pursuit of parenthood. Ultimately, the ethical landscape surrounding immunotherapies like IVIG and ILET is not black and white. It is defined by the relationship between scientific fact and compassionate care, and between protecting patients and honoring their choices. Navigating this space requires a delicate balance between clinical integrity and reproductive autonomy, one that acknowledges both the vulnerabilities and the decision-making capacities of patients who place extraordinary trust in the promise of medicine.

This paper examines the emerging role of immunological dysfunction in recurrent implantation failure (RIF) and recurrent pregnancy loss (RPL), analyzing both the biological mechanisms that impair maternal-fetal tolerance and the bioethical complexities surrounding the use of immunomodulatory treatments such as intravenous immunoglobulin (IVIG) and intralipid therapy. It argues that while these interventions offer hope to patients for whom conventional reproductive therapies have failed, their experimental nature raises critical concerns regarding informed consent, therapeutic misconception, and equitable access, requiring a careful balance between respecting reproductive autonomy and ensuring ethically responsible clinical care.

## Overview of the Maternal Immune System in Pregnancy

Pregnancy is far more than a passive biological state, it is an immunologically dynamic process that relies on a balance between the endocrine system, reproductive structures, and maternal immunity. While normally regarded as a natural and harmonious period of gestation, pregnancy from an immunological perspective is challenging: the maternal body must sustain a fetus that is genetically half foreign, yet

avoid mounting an immune response against it. The successful progression of pregnancy is therefore dependent upon a precisely regulated immune environment at both the systemic level and at the maternal-fetal interface. Fertilization marks the initiation of this process, where the fusion of an egg and a sperm results in a zygote that travels through the fallopian tube and eventually implants into the endometrium of the uterus (Kelleher et al., 2019). Implantation sets the stage for a highly specialized immunological environment to emerge, composed of fetal-derived trophoblasts and maternal decidual immune cells (DICs), including natural killer (NK) cells, macrophages, and T cells (Zhang et al., 2023). These immune cells do not simply tolerate the fetus; they actively participate in orchestrating the implantation, placental development, and protection of the fetus.

Trophoblasts, which form the outer layer of the developing placenta, play a foundational role in fetal survival. The placenta is the first organ created during development and it is the connection between a mother and a fetus (Staud & Karahoda, 2018). The outer layer is referred to as the trophoblast, which regulates functions such as digestive, excretory, immune, and more (Staud & Karahoda, 2018). Normally, cells in the body have specific markers called major histocompatibility complex (MHC) class I and II molecules, which alert the immune system to attack if the cells are harmful or if there is a foreign substance present (Staud & Karahoda, 2018). However, trophoblasts don't have these typical markers, so they don't trigger an immune response from the mother's body. Instead, they produce different MHC molecules known as [human leukocyte antigen] HLA-G and HLA-E, which are proteins found on the surface of most cells, and they help distinguish the body's own cells from foreign cells. They actually help prevent the mother's immune system from attacking the placenta (Altvater et al., 2021). Trophoblasts also release certain substances that influence nearby immune cells. For example, they can help immune cells called natural killer (NK) cells become less aggressive and instead support healthy blood vessel formation, which is important for the placenta to receive enough nutrients and oxygen (Staud & Karahoda, 2018).

These trophoblasts also help other immune cells called macrophages to shift into a more relaxed state during pregnancy, further ensuring that the immune system doesn't harm the developing fetus. In early pregnancy, macrophages are important for removing dead cells and helping tissues grow and repair, which is necessary for the developing placenta (True et al., 2022). The trophoblasts guide these macrophages to do this by producing molecules that help control their behavior, promoting a peaceful immune environment that supports pregnancy. Decidual macrophages (DMs) specifically assist in maintaining a healthy pregnancy by improving maternal-fetal tolerance, and interact with T-cells to prevent inflammation and immune responses from attacking the fetus. Another central player is the regulatory T cell (Treg) population. These cells expand significantly during pregnancy and are critical for maintaining tolerance to fetal antigens. Tregs suppress pro-inflammatory responses, modulate natural killer cell (NK) activity, and contribute to uterine structure. Experimental studies have shown that depletion of Tregs leads to increased risk of cytotoxic activity and fetal loss (Zhang et al., 2023).

Beyond immune cells, emerging research has highlighted the role of the uterine microbiota in shaping immune homeostasis during pregnancy. While once considered a sterile environment, the uterus and placenta host a community of commensal microbes that interact with trophoblasts and immune cells. These microbes can induce responses in trophoblasts through TLR4 signaling, leading to promotion of antiviral defense and immune response signaling (Zhang et al., 2023 & Hussain et al., 2022). However, too much of this immune response can be harmful, highlighting the importance of keeping the immune system in balance for a healthy pregnancy. Problems such as an overactive immune response or an imbalance in specific immune cells are increasingly being recognized as causes of these pregnancy issues, especially when more obvious factors like genetic or physical problems are ruled out.

Disruptions in these delicate immunological relationships can lead to reproductive complications such as recurrent implantation failure (RIF) and recurrent pregnancy loss (RPL). Abnormal activation of NK cells, deficiencies in Treg populations, and imbalances in cytokine profiles (e.g., elevated TNF-α or

IFN-γ relative to IL-10) have all been implicated in failed pregnancies (Hussain et al., 2022). When traditional causes such as genetic or anatomical abnormalities are ruled out, immune dysfunction is increasingly recognized as a leading contributor to unexplained reproductive failure. Understanding the immunological foundation of pregnancy transforms our perception of it from a passive state to an actively negotiated immunological event. It also opens the door for immune-targeted diagnostics and therapies, though these must be pursued with caution and backing by science.

***The Fetal-Maternal Interface and Immune Privilege***

The fetal-maternal interface is a highly specialized immunological site that enables the development of a genetically distinct fetus within the maternal body while minimizing the risk of immune rejection. As the immunological barrier and bridge between mother and fetus, primarily established at the placenta, is central to maintaining immune tolerance without compromising the body's ability to protect against pathogens (Zhang et al., 2023). It is here that immunological adaptation unfolds, as the maternal immune system balances two conflicting roles: defending against infection and accepting the semi-allogeneic fetus. From the moment of implantation, maternal tissues come into direct contact with fetal-derived trophoblasts, initiating a cascade of immune modulation. These trophoblast cells, which form the outer layer of the developing placenta, express unique molecules such as HLA-G and HLA-E rather than the classical MHC I and II antigens. These non-classical MHC molecules serve as tolerogenic signals that inhibit the cytotoxic activity of natural killer (NK) cells and cytotoxic T lymphocytes (Zhang et al., 2023). By doing so, trophoblasts help create a protected microenvironment where fetal antigens do not provoke an immune attack.

At the cellular level, several key immune populations converge at the decidua, the maternal portion of the placenta, to regulate this immune balance. Decidual natural killer cells (dNK), which comprise up to 70% of local immune cells in early pregnancy, differ markedly from their peripheral counterparts. Rather than mounting aggressive cytotoxic responses, dNK cells support fetal development by secreting vascular endothelial growth factor (VEGF), which promotes spiral artery remodeling and improves placental perfusion (Zhang et al., 2023). Decidual macrophages also undergo functional specialization. These decidual macrophages (DMs) can shift between phenotypes depending on local signals and the period of gestation the fetus is in (Zhang et al., 2023). As pregnancy progresses, they shift toward a phenotype characterized by increased phagocytic activity and the secretion of immunosuppressive cytokines, both of which further support maternal-fetal tolerance (Zhang et al., 2023). Regulatory T cells (Tregs) serve as a central regulatory force at the fetal-maternal interface. These T cells suppress inflammatory immune responses and facilitate tolerance to fetal alloantigens. Their numbers significantly increase during pregnancy and peak during the second trimester, which coincides with maximal trophoblast invasion and placental development (Huang, et al., 2020). Tregs exert their effects via both cell-contact dependent mechanisms and through the secretion of cytokines, which inhibit effector T cells and promote anti-inflammatory responses (Huang et al., 2020). Moreover, hormonal signals during pregnancy further modulate the immune response. Estrogen, progesterone, and human chorionic gonadotropin (hCG) have all been shown to enhance Treg proliferation and function. Estrogen increases Foxp3 expression, while progesterone promotes Treg expansion via glucocorticoid receptor signaling, and hCG not only augments Treg numbers but also facilitates their migration to the decidua (Huang et al., 2020). These hormonal effects ensure a systemic shift toward immune tolerance that supports the pregnancy as it advances.

The maternal-fetal interface also hosts an emerging third player: the uterine microbiota. Because the uterus harbors microbes, it can influence the development of immune regulation and promote the expression of interferon-stimulated genes (ISGs) (genes from the innate immune response), and thereby modulate the local immune milieu in favor of tolerance and pathogen defense (Zhang et al., 2023). When this balance is disrupted, the consequences can be severe. However, recurrent implantation failure (RIF)

and recurrent pregnancy loss (RPL) often arise from immune dysregulation. Studies have implicated overactivation of NK cells and a deficit of Tregs in the pathophysiology of these conditions (Huang et al., 2020 & Zhang et al., 2023). The failure to establish proper immune tolerance at the fetal-maternal interface may lead to embryo rejection, poor placental development, or chronic inflammation, all of which contribute to reproductive failure.

***The Role of Immune Cells in the Support and Rejection of the Fetus***

Immune cells, including natural killer (NK) cells, regulatory T (Treg) cells, macrophages, and dendritic cells, all play pivotal roles in ensuring this balance is maintained. When these immune responses are disrupted, it can lead to recurrent implantation failure (RIF) and recurrent pregnancy loss (RPL). One of the most significant immune cell populations at the maternal-fetal interface are NK cells. They are crucial for the remodeling of maternal blood vessels, a process essential for the development of the placenta and adequate nutrient exchange between mother and fetus (Ma et al., 2023). Decidual NK cells are unique in that they exhibit reduced cytotoxicity compared to peripheral NK cells. This is critical because the maternal immune system must recognize and interact with the semi-allogeneic fetus without causing harm. Unlike typical NK cells, decidual NK cells do not exhibit the full cytotoxic activity against fetal extravillous trophoblast (EVT) cells. Instead, they secrete growth factors such as vascular endothelial growth factor (VEGF) and angiopoietins, which promote angiogenesis and contribute to the remodeling of spiral arteries. These factors are essential for ensuring proper blood flow to the placenta (Ma et al., 2023).

However, NK cell dysfunction is a key factor in several pregnancy complications, particularly in RIF and RPL. Abnormal NK cell activation, such as an increased cytotoxic phenotype or an imbalance in activating and inhibitory signals, can lead to impaired vascular remodeling and reduced implantation rates. In fact, a reduced number of decidual NK cells or abnormal activity of these cells is often associated with failed pregnancies (Balasundaram & Farhana, 2025). Furthermore, the genetic makeup of the maternal NK cell receptors, particularly the killer-cell immunoglobulin-like receptors (KIRs), can influence pregnancy outcomes. A specific KIR and HLA-C genotype combination has been linked to a higher risk of pregnancy complications such as preeclampsia, and women with a certain KIR-AA genotype may be more susceptible to RIF due to poor trophoblast invasion and impaired vascular remodeling (Ma et al., 2023). Another critical immune cell population involved in maintaining pregnancy is regulatory T (Treg) cells. These cells are responsible for ensuring that the maternal immune system does not attack the fetus, which is genetically different from the mother. Treg cells promote immune tolerance by suppressing the activity of other immune cells. Treg cells are found in the decidua, where they help to modulate immune responses during pregnancy. Their role is to prevent maternal immune rejection of the fetus and to support the necessary immune adaptation to pregnancy (Balasundaram & Farhana, 2025). The proper function of Treg cells is essential for a successful pregnancy, as their dysfunction can lead to an overactive immune response and an increased risk of pregnancy failure. In cases of RPL or RIF, an imbalance between Treg cells and other immune cell populations, such as Th1 or Th17 cells, can result in excessive inflammation and immune rejection of the fetus. This can lead to failed implantation, miscarriage, or complications such as preeclampsia. Reduced numbers or impaired function of Treg cells are often observed in women with immune-mediated pregnancy loss (Ma et al., 2023). Interestingly, therapies aimed at increasing the number and function of Treg cells, such as intravenous immunoglobulin (IVIG) or other immune-modulating treatments, have shown promise in improving pregnancy outcomes for women with RPL or RIF (Ma et al., 2023).

Macrophages are another key immune cell population at the maternal-fetal interface. These cells are involved in a wide range of functions, including tissue remodeling, phagocytosis of apoptotic cells, and immune regulation. Macrophages in the decidua are crucial for maintaining a healthy pregnancy, as they help with spiral artery remodeling, trophoblast invasion, and the maintenance of immune tolerance.

They exist in two primary forms: pro-inflammatory (M1) and anti-inflammatory (M2). During pregnancy, the M2 macrophage phenotype predominates and plays a key role in promoting tissue remodeling and immune tolerance (Balasundaram & Farhana, 2025). In women with RPL or RIF, macrophage dysfunction can contribute to implantation failure and pregnancy loss. For example, an increased number of M1 macrophages and a shift towards a pro-inflammatory state can disrupt the remodeling of spiral arteries, impair trophoblast invasion, and contribute to placental insufficiency (Ma et al., 2023). Additionally, macrophages at the maternal-fetal interface secrete cytokines like IL-10 and TGF-β, which help to suppress inflammation and promote immune tolerance, thus supporting fetal development. An imbalance in these cytokines or an excess of pro-inflammatory macrophages can disrupt pregnancy and lead to complications such as preeclampsia or miscarriage.

Dendritic cells (DCs), although a smaller population in the decidua compared to NK cells and macrophages, play a critical role in initiating and regulating immune responses during pregnancy. DCs are antigen-presenting cells that recognize and process fetal antigens, thus playing a crucial role in maternal-fetal immune interactions. They help to promote maternal immune tolerance and are involved in the modulation of T cell responses at the maternal-fetal interface. Abnormalities in the function of DCs, particularly an increased number of mature DCs and a reduced number of immature DCs, have been linked to adverse pregnancy outcomes such as RPL and implantation failure (Ma et al., 2023). DCs are also involved in the modulation of the immune environment in the decidua by influencing the differentiation of other immune cells, such as Treg cells and NK cells. In women with RIF, an increase in mature DCs has been associated with impaired implantation, suggesting that an overactive immune response may hinder the delicate balance required for pregnancy maintenance (Ma et al., 2023). Therefore, understanding the function and regulation of DCs at the maternal-fetal interface is crucial for identifying potential therapeutic targets to improve implantation success in women with RIF.

Decidual NK cells, Treg cells, macrophages, and dendritic cells all contribute to this delicate balance, and any dysfunction in these cells can lead to pregnancy complications such as RPL and RIF. Understanding the roles of these immune cells and their interactions with trophoblasts and other maternal cells is essential for developing new diagnostic and therapeutic strategies for women experiencing immune-mediated infertility. Ongoing research into the immunological foundation of pregnancy will help refine our understanding of the complex immune processes involved in pregnancy and provide new avenues for treatment and prevention of pregnancy loss.

## Logistics and Mechanisms of Immunotherapies in Reproductive Medicine

In the context of recurrent pregnancy loss (RPL) and recurrent implantation failure (RIF), immunological disturbances at the maternal-fetal interface are increasingly recognized as critical contributors to reproductive failure. Two key therapeutic strategies, intravenous immunoglobulin (IVIG) and intralipid therapy, have been introduced to mitigate these disruptions by directly modulating immune cell function, cytokine signaling, and tolerance mechanisms.

IVIG is a plasma-derived product composed predominantly of pooled IgG antibodies. It operates on multiple immunological levels to restore immune tolerance in women with suspected immune-mediated reproductive complications. A central mechanism of IVIG is the suppression of natural killer (NK) cell cytotoxicity, which is often elevated in patients with RPL or RIF. IVIG can bind to Fcγ receptors (surface proteins that bind to antibodies) on NK cells, thereby blocking their activation and cytolytic function (Garmendia et al., 2025). This effect reduces the likelihood of NK cell-mediated damage to trophoblasts, promoting a more stable environment for embryo implantation and placental development. Another vital mechanism involves the expansion of regulatory T cells (Tregs). Tregs are essential for suppressing maternal immune responses to paternal antigens and maintaining fetal tolerance. IVIG enhances both the number and function of Tregs, contributing to a restored immune balance at the implantation site (Garmendia et al., 2025; Campos Canella et al., 2021). This expansion of

Tregs also inhibits the activity of inflammatory cells, further supporting fetal-maternal immune harmony. By decreasing levels of inflammatory cytokines while also promoting anti-inflammatory cytokines, IVIG therapy creates an immune balance conducive to successful pregnancy outcomes (Garmendia et al., 2025). Moreover, IVIG can neutralize pathogenic antibodies and inhibit complement activation (system of proteins activating to eliminate pathogens), both of which are implicated in some autoimmune forms of pregnancy loss. By engaging with Fc receptors on antigen-presenting cells and modulating their activation, IVIG prevents downstream activation of effector T cells and minimizes inflammation at the maternal-fetal interface (Garmendia et al., 2025). Together, these mechanisms contribute to the suppression of embryo-targeted immune responses, ultimately enhancing the likelihood of implantation and sustained pregnancy.

Intralipid therapy involves intravenous infusion of a fat emulsion composed of soybean oil, egg phospholipids, and glycerin. Originally developed for nutritional support, intralipid has demonstrated immunomodulatory effects relevant to reproductive medicine. The primary action of intralipid is its ability to dampen abnormal NK cell activity. In women with RPL and RIF, heightened NK cytotoxicity is frequently observed. Intralipid therapy reduces the cytolytic potential of NK cells, particularly by influencing receptor-mediated signaling and intracellular pathways involved in cell activation (Campos Canella et al., 2021). By inhibiting this cytotoxicity, intralipid therapy helps protect trophoblast cells during the critical window of implantation. Additionally, intralipid influences cytokine signaling and (T-helper cells) Th1/Th2 balance. Similar to IVIG, intralipid therapy has been associated with reductions in pro-inflammatory cytokines and elevations in cytokines that support implantation (Campos Canella et al., 2021). These cytokine shifts may facilitate embryo adhesion, placental development, and immune tolerance. Furthermore, intralipid may contribute to an improved decidual immune environment. Its fatty acid components are thought to interact with dendritic cells and macrophages, encouraging them to adopt an anti-inflammatory or M2-like phenotype (Zhang et al., 2023). This shift in macrophage function supports tissue remodeling and angiogenesis, both of which are necessary for successful implantation and placentation. Unlike IVIG, intralipid does not directly expand Treg populations but exerts indirect immunosuppressive effects by decreasing inflammatory stress and modulating innate immune cell responses (Zhang et al., 2023). These changes may permit regulatory pathways to operate more effectively, contributing to the overall tolerogenic shift required for pregnancy continuation.

Both IVIG and intralipid therapies exert multifaceted effects on the immune system, targeting both innate (NK cells, macrophages) and adaptive (T cells, cytokine production) immune components. IVIG acts primarily through antibody-mediated regulation, promoting Treg expansion, Fc receptor blockade, and cytokine profile shifts. In contrast, intralipid therapy modulates cell signaling and cytokine secretion via lipid-based pathways, particularly affecting NK cell activity. By recalibrating immune responses at the maternal-fetal interface, these therapies offer immunological support to women with immune-mediated forms of RPL and RIF, and their mechanistic actions provide the scientific rationale for ongoing clinical use and investigation (Campos Canella et al., 2021; Garmendia et al., 2025).

### *Evidence For and Against Intravenous Immunoglobulin (IVIG) Therapy*

Intravenous immunoglobulin (IVIG) therapy has emerged as a potential treatment for women with recurrent pregnancy loss (RPL) or recurrent implantation failure (RIF), particularly in cases where these conditions are believed to be immune-mediated. IVIG works by modulating the immune system, suppressing natural killer (NK) cell activity, promoting regulatory T cells (Tregs), neutralizing autoantibodies, and reducing pro-inflammatory cytokines, all of which are critical in pregnancy maintenance (Yamada et al., 2022). These immune modulatory effects are beneficial, especially in women with abnormal immune responses that might interfere with successful pregnancy, such as elevated NK cell activity and pro-inflammatory cytokine production.

Several studies have shown promising results regarding the efficacy of IVIG in improving pregnancy outcomes. For instance, a recent double-blind, placebo-controlled trial by Yamada et al. (2022) demonstrated that IVIG treatment significantly increased the ongoing pregnancy rate at 22 weeks of gestation and live birth rates in women with unexplained RPL, particularly in those who received treatment in the early stages of pregnancy (4–5 weeks gestation). This study found that IVIG therapy significantly reduced the risk of miscarriage in women with a history of multiple pregnancy losses, with ongoing pregnancies at 22 weeks reaching 62% in the IVIG group compared to 34% in the placebo group (Yamada et al., 2022). These findings suggest that IVIG can be highly beneficial for women with immune-mediated RPL, providing hope to those who have struggled with repeated pregnancy losses. However, despite these positive results, IVIG therapy has notable drawbacks. One key concern is the variability in treatment response. While the Yamada study showed improved outcomes for early-stage treatments, the efficacy was less clear in cases where treatment began later in pregnancy (around 6 weeks gestation), highlighting the importance of timely intervention (Yamada et al., 2022). Moreover, larger trials and meta-analyses have reported mixed results, with some studies failing to show significant improvements in pregnancy outcomes with IVIG compared to placebo, suggesting that the benefits of IVIG are not universally applicable (Christiansen et al., 2019).

Additionally, IVIG is an expensive treatment, and its high cost is not always covered by insurance, which can limit its accessibility for many women. Side effects are also a concern, including allergic reactions, infusion-related side effects such as fever and headache, and more serious complications like thromboembolism and renal dysfunction (Yamada et al., 2022). Furthermore, the treatment is not without risks for the fetus, as the rates of preterm delivery and fetal growth restriction were higher in the IVIG group compared to the placebo group in the study, indicating potential complications despite its immune-modulatory effects (Yamada et al., 2022). IVIG therapy presents a promising treatment for women with immune-related recurrent pregnancy loss, especially when administered early in pregnancy. However, its high cost, potential side effects, and inconsistent efficacy across studies call for cautious and individualized consideration in clinical practice. Further large-scale studies are needed to better define the treatment protocols and ensure its safety and effectiveness for pregnant women with RPL.

### *Evidence For and Against Intralipid Therapy*

Intralipid therapy (or lipid emulsion therapy, LET) is an emerging treatment in reproductive medicine aimed at managing immune-mediated infertility, including recurrent pregnancy loss (RPL) and recurrent implantation failure (RIF). The therapy involves intravenous administration of a fat-based emulsion composed of soybean oil, egg phospholipids, and glycerin. Originally developed for parenteral nutrition, intralipid has been repurposed to modulate immune function, primarily by suppressing natural killer (NK) cell activity, promoting regulatory T cell (Treg) function, and modulating pro-inflammatory cytokine production in the uterine environment (Kumar et al., 2021).

Advocates of intralipid therapy point to its potential to improve pregnancy outcomes in women with immune-mediated reproductive failure. Several studies and clinical observations suggest that intralipid can downregulate cytotoxic NK cell activity, which is often elevated in women with RPL and RIF. Additionally, intralipid therapy has been shown to increase the function of Tregs, which are crucial for maintaining immune tolerance at the maternal-fetal interface (Kumar et al., 2021). By reducing the secretion of pro-inflammatory cytokines such as TNF-α and IFN-γ, intralipid helps create a more favorable immune environment for pregnancy. Meta-analyses of several studies indicate that intralipid therapy significantly improves clinical pregnancy rates (CPR), live birth rates (LBR), and implantation rates (IR), while simultaneously reducing miscarriage rates (MR). For example, a meta-analysis by Kumar et al. (2021) found that intralipid therapy increased LBR by a factor of 2.36 and reduced MR by 80%. These findings suggest that intralipid therapy can be particularly beneficial for women with unexplained infertility or immune dysfunction, particularly in those whose previous treatments have failed.

Yet despite the promising outcomes in some studies, the evidence base for intralipid therapy remains limited and largely anecdotal. A significant concern is the lack of large, well-controlled randomized controlled trials (RCTs), with many studies being open-label or lacking placebo controls. This absence of robust evidence makes it difficult to draw definitive conclusions about the therapy's effectiveness. Furthermore, the lack of standardization in immune testing protocols complicates the interpretation of results. What one clinic considers "elevated NK activity" may differ from another clinic's definition, making it challenging to identify the specific patient populations most likely to benefit from intralipid therapy (Kumar et al., 2021). While intralipid is generally well-tolerated and much less expensive than alternatives like intravenous immunoglobulin (IVIG), it can still cause side effects. These may include allergic reactions, particularly in patients with sensitivities to egg or soy proteins, which are key components of the emulsion. Moreover, the long-term safety of intralipid therapy in reproductive contexts is not well-established, and further studies are needed to understand any potential risks fully. Another concern is that intralipid therapy is not universally accepted as a routine intervention for RPL or RIF. The effectiveness of the therapy appears to be limited to specific patient groups, particularly those with identified immunological risk factors (Kumar et al., 2021). As such, intralipid should be considered more as an experimental treatment rather than a standard protocol.

Overall, intralipid therapy presents a promising alternative for women with immune-mediated reproductive failure, particularly when standard treatments have failed. While the therapy has shown beneficial effects in several studies, including improvements in pregnancy outcomes and a reduction in miscarriage rates, the overall evidence remains inconclusive due to the lack of large, rigorous trials. As such, intralipid therapy should be considered on a case-by-case basis, with careful attention to patient selection and potential risks. More high-quality studies are needed to establish its efficacy and safety in reproductive medicine. This is where the issue of healthcare provider-patient disclosure and trust comes in.

## Bioethical Analysis of Reproductive Autonomy, Informed Consent, and Accessibility

Reproductive autonomy is a fundamental concept in medical ethics, emphasizing the right of individuals to make independent and informed decisions about their reproductive health. It refers to a woman's ability to make decisions regarding contraception, pregnancy, and childbearing without undue interference or coercion (Upadhyay et al., 2014). This autonomy is central to the idea of personal freedom and dignity in reproductive decision-making. In ethical discussions surrounding reproductive autonomy, it is essential to recognize that autonomy involves more than just the ability to make decisions; it also includes the capacity to make decisions free from coercion, manipulation, or external pressures, particularly from partners, family members, or societal norms (Upadhyay et al., 2014). In reproductive contexts, autonomy means the ability for a woman to decide if, when, and how to become pregnant, to use contraception, and to choose whether to continue a pregnancy. These decisions are often influenced by factors such as gender roles, cultural expectations, and the level of control a woman has in her relationship and societal environment (Upadhyay et al., 2014). For women facing infertility or recurrent pregnancy loss (RPL), the stakes are especially high, as these individuals may feel immense pressure to pursue various treatment options, even if those treatments may be experimental or carry risks.

Informed consent is a cornerstone of ethical medical practice, ensuring that patients fully understand the nature of the treatments being offered, including the potential risks, benefits, and uncertainties involved. In the context of complex and uncertain treatments such as those used for recurrent pregnancy loss (RPL) and recurrent implantation failure (RIF), informed consent becomes particularly challenging. The key elements of informed consent include providing patients with comprehensive, understandable information, ensuring that the decision is made voluntarily, and confirming that patients have the capacity to make decisions based on the information provided (Nijhawan et al., 2013). For treatments like intravenous immunoglobulin (IVIG) or intralipid therapy, both

of which are experimental therapies for women with immune-mediated infertility, the risks and benefits may not be fully understood. Despite the lack of robust evidence for these treatments' efficacy, patients are often presented with these options due to the desperation and emotional vulnerability that accompany recurrent miscarriage and infertility. The ethical implications of offering such treatments are profound, as patients might feel pressured to consent to therapies that may not be the best option or may carry unknown risks. Valid informed consent in this context must address not only the medical facts but also the emotional and psychological state of the patient. The decision to undergo experimental treatments may be influenced by the emotional desire to have a child, potentially clouding a patient's ability to make a fully informed decision. As Nijhawan et al. (2013) point out, informed consent is not simply a formality, it is an ongoing process that requires clear communication and consideration of the patient's emotional and psychological vulnerabilities.

The pressure to pursue "every possible treatment" is a significant ethical challenge in the context of RPL and RIF. As many patients facing infertility or repeated pregnancy losses may feel desperate for a solution, their decision-making autonomy can become compromised. For instance, patients may be presented with multiple experimental options, such as IVIG or intralipid therapy, even though the evidence for these treatments is often inconclusive or lacking in rigor. This can lead to situations where patients consent to treatments based on the hope of achieving pregnancy, even when the benefits and risks are not fully understood (Garmendia et al., 2025). Emotional vulnerability plays a key role in this phenomenon. Women experiencing RPL or RIF often face immense emotional stress, and the pressure to become pregnant can lead to compromised autonomy. They may feel that they have no choice but to pursue all available treatments, even if they are not fully informed about the uncertainties or potential harms of these treatments. Reproductive coercion is a real concern, especially when a patient's reproductive decisions are influenced by a desire to meet the expectations of family members or partners, rather than being based on fully autonomous decision-making (Upadhyay et al., 2014). This coercion can occur in various forms, such as emotional pressure or partner-driven decisions, further complicating the informed consent process.

Offering experimental treatments to patients who are emotionally vulnerable raises significant ethical questions. The ethical principle of beneficence, the obligation to do good, requires that healthcare providers offer treatments that are truly beneficial to patients. However, when the evidence for a treatment's efficacy is uncertain, offering such treatments may not align with the principle of beneficence, as it could expose patients to potential harm or false hope (Nijhawan et al., 2013). Additionally, providing experimental options like IVIG or intralipid therapy to patients with RPL or RIF without adequate evidence of efficacy may also violate the principle of non-maleficence, which dictates that healthcare providers must avoid causing harm. Patients who are desperate to conceive may be influenced by the emotional and psychological weight of their situation, leading them to consent to treatments without fully understanding the risks involved. This dynamic can compromise their ability to exercise genuine autonomy in making decisions about their healthcare (Garmendia et al., 2025).

Reproductive autonomy and informed consent are crucial elements in medical ethics, particularly when dealing with complex and uncertain treatments for conditions like RPL and RIF. Ensuring that patients fully understand the risks, benefits, and uncertainties of treatments like IVIG and intralipid therapy requires a careful and sensitive approach that takes into account not only the medical facts but also the emotional and psychological context in which these decisions are made. Healthcare providers must be mindful of the potential for coercion and emotional vulnerability and must ensure that patients are making fully informed decisions based on their own values and goals, rather than external pressures or unrealistic hopes.

### *Experimental Use and the Ethics of Hope vs. Harm*

The therapeutic misconception occurs when patients participating in clinical trials, particularly those involving experimental treatments for conditions such as recurrent pregnancy loss (RPL) or recurrent implantation failure (RIF), fail to distinguish between the goals of clinical research and those of routine medical care. Often, patients assume that participation in a clinical trial is intended to offer them the best possible individualized treatment, which may not necessarily align with the research objectives (Lidz & Appelbaum, 2002). This misconception is particularly problematic in reproductive medicine, where the emotional toll of infertility or recurrent pregnancy loss can cloud a patient's judgment. As patients become desperate for solutions, they may overestimate the potential benefits of unproven therapies, such as intravenous immunoglobulin (IVIG) or intralipid therapy, believing these treatments are a guaranteed path to success, even when scientific evidence supporting their efficacy is limited. The role of hope in this therapeutic misconception cannot be overstated. Patients often enter fertility treatments with high expectations, driven by the hope of achieving pregnancy, which can lead them to consent to experimental therapies without fully understanding the risks, uncertainties, or the lack of conclusive evidence regarding the efficacy of these treatments. The ethical challenge here lies in the conflict between the therapeutic intent of healthcare providers, who are bound by the principles of beneficence and non-maleficence, and the reality of experimental treatments, which may not offer immediate, tangible benefits to patients. When patients' hopes for a successful pregnancy cloud their understanding of the experimental nature of these therapies, their ability to make fully informed decisions is compromised.

Fertility clinics play a crucial role in the landscape of reproductive medicine, especially in the context of experimental treatments for RPL and RIF. However, the ethical responsibility of these clinics is complex. While clinics have a duty to provide the best available care, there are growing concerns regarding the promotion of unproven therapies, especially those that are not supported by rigorous scientific evidence. The increasing use of off-label drugs and treatments, such as IVIG and intralipid therapy, in private fertility clinics is an example of this issue (Usadi & Merriam, 2015). These treatments often promise hope, but in many cases, they lack the large-scale clinical trials or long-term safety data needed to substantiate their use. The commercialization of fertility treatments also poses significant ethical challenges. In some cases, fertility clinics may offer unproven or experimental therapies as part of their treatment protocols, not necessarily because they have been shown to be effective, but because they cater to patients' desires for "last-resort" options. This practice may exacerbate the therapeutic misconception, as patients, particularly those who are desperate to conceive, are offered treatments that promise success but may not have a reliable track record. In such environments, the clinic may prioritize profit over patient autonomy, leading to ethical dilemmas related to informed consent and the provision of treatments that may ultimately do more harm than good. Clinicians in these settings must balance their duty to provide compassionate care with the responsibility to ensure that patients fully understand the experimental nature of treatments. Providing clear, evidence-based information is essential, yet in the competitive world of fertility clinics, there may be a tendency to oversell therapies that have not been rigorously tested or validated. The ethical responsibility of clinicians to present treatments honestly, along with a discussion of the risks and uncertainties, is critical to maintaining the integrity of the informed consent process (Usadi & Merriam, 2015).

Clinicians who offer experimental treatments for conditions like RPL or RIF have an ethical duty to clarify the benefits and risks of these treatments to their patients. This responsibility is magnified when dealing with therapies that are not well-supported by clinical evidence. A clinician's duty is not only to offer medical interventions but to ensure that patients understand the potential outcomes of their choices (Usadi & Merriam, 2015). In the case of treatments such as IVIG or intralipid therapy, which have shown mixed results in clinical studies, it is crucial that clinicians clearly communicate the lack of conclusive evidence supporting these therapies. A fundamental aspect of this ethical responsibility is ensuring that the patient's decision to pursue experimental treatments are informed and voluntary,

rather than driven by hope or emotional desperation. As discussed in the context of the therapeutic misconception, patients facing infertility or recurrent pregnancy loss may have a strong emotional drive to pursue "every possible treatment," which can cloud their understanding of the risks associated with such therapies. By emphasizing the experimental nature of these treatments and discussing alternative options, clinicians can help mitigate the therapeutic misconception and support the patient in making an informed, autonomous decision about their care.

Moreover, clinicians should be transparent about the potential for harm. In some cases, experimental therapies may carry risks that are not fully understood, either due to a lack of long-term data or because they are administered in ways that have not been rigorously studied. For example, while IVIG has shown promise in some small studies, it is expensive, carries risks of allergic reactions, and may not always lead to successful pregnancy outcomes. Clinicians should engage in an open dialogue with patients about these risks, helping them weigh the benefits against the potential harm. This transparency fosters trust and ensures that patients' decisions are made based on a clear understanding of the treatment's scientific foundation.

"Compassionate use" refers to the provision of experimental treatments outside of formal clinical trials, typically for patients who have exhausted all other options (Lidz & Appelbaum, 2002). This practice raises important ethical considerations, particularly in the context of experimental immunotherapies used in reproductive medicine, such as IVIG or intralipid therapy. While compassionate use may provide hope to patients who are desperate for a solution, it also introduces risks related to the lack of comprehensive safety data. The ethical challenge with compassionate use lies in the tension between the desire to provide care and the necessity of protecting patients from potential harm. Offering experimental treatments without solid evidence of their efficacy risks exposing patients to treatments that may not only fail to help but may also cause unforeseen complications. For example, although IVIG has been used for immune-mediated infertility, its high cost and the uncertainty surrounding its effectiveness raise questions about whether it is appropriate to offer it as a compassionate use treatment. Similarly, intralipid therapy, while promising, has not undergone large-scale, randomized trials to establish its safety and efficacy for treating RPL or RIF, making it ethically controversial to offer it outside of a clinical trial context (Usadi & Merriam, 2015). The ethical principle of beneficence mandates that clinicians act in the best interest of their patients, but in the case of compassionate use, this principle can conflict with the obligation to minimize harm (non-maleficence). The lack of rigorous evidence for these therapies means that their provision may not align with the patient's best interest, raising concerns about the ethical implications of offering unproven treatments. Clinicians must carefully consider whether the potential benefits of offering these therapies outweigh the risks and whether it is more appropriate to refer patients to clinical trials where the treatments can be studied under controlled conditions.

Overall, the use of experimental therapies for RPL and RIF is fraught with ethical dilemmas. The therapeutic misconception, the promotion of unproven therapies by fertility clinics, and the ethical responsibility of clinicians to clarify the risks and benefits of experimental treatments all pose challenges to informed consent. While the desire to provide hope to patients facing infertility or recurrent pregnancy loss is understandable, clinicians must be mindful of the potential harm caused by offering unproven treatments without adequate evidence. Ethical practice requires transparency, clear communication, and a commitment to ensuring that patients make truly informed decisions, free from the influence of hope-driven misconceptions or external pressures.

### *Maternal vs. Fetal Ethical Interests*

The use of immunosuppressive treatments during pregnancy poses ethical concerns when maternal health requires treatment that may simultaneously harm the fetus. Immunosuppression is essential for women who require organ transplants or suffer from autoimmune disorders, as these treatments prevent

rejection and control disease activity. However, these medications carry the risk of adverse fetal outcomes, including congenital malformations, intrauterine growth restriction, and even fetal death, especially when used during critical periods of fetal development (Armenti et al., 2002). Immunosuppressive drugs such as corticosteroids, azathioprine, cyclosporine, and tacrolimus have been shown to have varying levels of teratogenicity, with some posing greater risks than others (Armenti et al., 2002). The challenge lies in balancing the therapeutic need for maternal health with the potential harms to the fetus, a situation that complicates the ethical decision-making process. In these cases, healthcare providers must carefully consider both the necessity of immunosuppression to ensure the mother's survival and the risk posed to the developing fetus. The risks to the fetus may be mitigated by adjusting medication dosages or switching to less teratogenic alternatives when feasible (Armenti et al., 2002).

The dilemma becomes even more complex when the treatment necessary for fetal well-being may compromise the mother's health. An example of this occurs when immunosuppressive treatments, such as high-dose corticosteroids or tacrolimus, are used to prevent immune-mediated pregnancy complications but lead to significant maternal health risks, including the development of diabetes, hypertension, or even long-term kidney dysfunction (Armenti et al., 2002). This scenario poses significant ethical challenges as maternal health may deteriorate while the fetus remains at risk. Physicians must balance the dual responsibility of promoting fetal health while ensuring that the treatments do not endanger the mother's long-term health. In cases where immunosuppressive therapy is essential for preventing rejection or managing autoimmune diseases, clinicians must navigate these complex decisions with great care, ensuring that the risks to the mother are minimized and that the mother's autonomy is respected throughout the decision-making process.

Bioethical principles provide a framework for addressing the complex ethical concerns surrounding maternal-fetal treatment. Principlism, a widely used approach in medical ethics, comprises four main principles: autonomy, beneficence, non-maleficence, and justice (Beauchamp & Childress, 2019). Each of these principles must be carefully applied to balance the competing interests of the mother and the fetus. Autonomy is a principle that emphasizes the right of the pregnant woman to make decisions regarding her health and treatment. In the context of immunosuppressive therapy, the woman's autonomy must be respected, particularly when making informed decisions about whether to proceed with treatments that may have unknown or harmful effects on the fetus. Autonomy also entails that the patient has access to all necessary information regarding the risks and benefits of treatment options, enabling her to make decisions that align with her personal values and desires. Beneficence highlights the duty of healthcare providers to act in the best interests of their patients. In cases of maternal-fetal conflict, beneficence demands that clinicians work toward the well-being of both the mother and the fetus. When maternal treatment may harm the fetus, the principle of beneficence requires physicians to choose interventions that optimize health outcomes for both parties. For instance, when selecting immunosuppressive drugs, a clinician must weigh the maternal benefits of preventing disease exacerbation or rejection against the fetal risks of potential malformations.

Non-maleficence, or the principle of "do no harm," emphasizes the need to avoid causing harm to both the mother and the fetus. In scenarios where immunosuppressive treatment is necessary, it is crucial that healthcare providers minimize the risks of teratogenic effects or fetal toxicity. This principle calls for the use of the least harmful treatment options and the consideration of alternatives when available. Finally, justice is a principle related to fairness in the distribution of healthcare resources and ensuring equitable access to treatments. In the context of pregnancy and immunosuppressive therapy, justice requires that all patients, regardless of socioeconomic status or background, have access to evidence-based treatments that balance maternal and fetal health risks. Moreover, justice entails that clinicians provide equal care and ensure that decisions made in the interest of the mother are not influenced by bias or unequal access to medical resources.

While principlism provides a structured approach, relational autonomy offers a nuanced framework that considers the interconnectedness of individuals within social contexts (Saad, 2017). In the case of pregnant women, relational autonomy takes into account the influence of familial, societal, and cultural contexts on a woman's decision-making process. This perspective challenges the individualistic view of autonomy by acknowledging that decisions about maternal health and fetal well-being are often made within a web of relationships, including those with partners, family members, and healthcare providers. For instance, a woman may feel pressure from her partner or family members to pursue certain treatments, especially when pregnancy loss has occurred repeatedly. These external influences can sometimes overshadow a woman's individual decision-making power, making her less autonomous than she may appear. Relational autonomy recognizes these complexities and calls for a collaborative approach in decision-making, where healthcare providers actively engage with the woman and her support system to ensure that her decision is truly reflective of her values and wishes (Saad, 2017). In cases where maternal treatment may harm the fetus, relational autonomy emphasizes the importance of considering not just the woman's wishes but also her social context. It requires clinicians to facilitate open and honest conversations with patients, helping them to navigate the emotional, social, and medical complexities of their treatment decisions. Relational autonomy highlights the importance of creating a supportive environment where a woman's decision-making autonomy is respected, but her relationships and social context are also taken into consideration.

The ethical challenges surrounding maternal treatment that may harm the fetus, particularly in the context of immunosuppressive therapies, require careful balancing of competing interests. The guidelines of principlism (autonomy, beneficence, non-maleficence, and justice) provide a framework for making ethically sound decisions, but relational autonomy further enriches this framework by recognizing the importance of social context and interdependence. Ultimately, ethical decision-making in reproductive medicine must be guided by a deep respect for the patient's autonomy, a commitment to minimizing harm, and a recognition of the broader relational dynamics that shape healthcare choices. By adhering to these ethical principles, healthcare providers can navigate the difficult terrain of maternal-fetal treatment decisions in a manner that is both ethically responsible and compassionate.

### *Equity and Accessibility*

One of the most significant barriers to access in the field of reproductive immunology is the high cost associated with therapies like intravenous immunoglobulin (IVIG) and intralipid therapy. These treatments, often employed in the management of recurrent pregnancy loss (RPL) and recurrent implantation failure (RIF), are expensive and often not covered by insurance policies. This raises considerable ethical concerns, as the financial burden placed on patients seeking these therapies can limit access to care, particularly for those from lower socioeconomic backgrounds. As these treatments are often considered experimental, their coverage under insurance plans is sporadic, leaving many women who might benefit from these therapies unable to afford them. The commercialization of these therapies exacerbates the problem by positioning them as luxury treatments for those who can afford them, rather than as accessible medical interventions. This creates a disparity in access to care, as wealthier individuals are more likely to have the means to pursue these treatments, whereas others may have to forgo them due to financial constraints (Usadi & Merriam, 2015). In reproductive medicine, where the emotional and physical costs of infertility and pregnancy loss are already high, the addition of financial barriers to accessing potentially beneficial therapies raises profound ethical questions about fairness and justice in healthcare. The implications of these cost-related barriers extend beyond direct financial strain; they also affect patients' autonomy, as the lack of affordable options may force individuals to make decisions based on financial considerations rather than medical needs or personal desires. This situation is compounded by the lack of universal healthcare coverage for fertility

treatments, leaving many individuals to navigate a complex and often overwhelming landscape of financial and emotional challenges while trying to conceive (Røstvik, 2025).

The commercialization of reproductive immunology also raises significant ethical concerns, particularly in the context of treatments like IVIG and intralipid therapy, which are often marketed to women facing fertility challenges. Fertility clinics, in particular, may offer these therapies as part of their treatment packages, even when the scientific evidence supporting their efficacy is still inconclusive. The commercialization of these therapies can create a false sense of certainty among patients, who may be led to believe that these treatments are well-established and highly effective options, even when they are not fully supported by the scientific community (Usadi & Merriam, 2015). Fertility clinics may also capitalize on the emotional vulnerability of patients who are desperate to conceive, offering expensive therapies that may not be the most effective or necessary for their specific condition. This situation can be seen as an ethical conflict, where the profit-driven motives of commercialized healthcare providers may conflict with the best interests of the patient, who might be making decisions based on incomplete or biased information (Røstvik, 2025). Furthermore, the growing use of unregulated and costly treatments in the private fertility sector raises concerns about the lack of oversight in ensuring that these treatments are both safe and effective. While these therapies may provide a sense of hope for patients, the risks associated with their widespread use in the absence of rigorous scientific validation pose ethical challenges, especially when they are marketed without full disclosure of potential risks or uncertainties. Given the growing commercial demand for reproductive immunotherapies like IVIG and intralipid therapy, the question of whether these treatments should be regulated more tightly becomes a critical issue.

As fertility clinics and pharmaceutical companies continue to offer these treatments, there is a need for greater oversight to ensure that they are being provided based on solid evidence of efficacy and safety, rather than on unsubstantiated claims or profit motives. Currently, many of these treatments are not subject to the same level of regulation as other medical therapies. IVIG, for example, is used off-label for fertility purposes despite limited evidence of its effectiveness in treating RPL or RIF. Similarly, intralipid therapy has shown some promise, but its use remains controversial, and larger, more controlled studies are needed to fully understand its role in reproductive medicine (Koutras et al., 2022). The lack of consistent regulatory oversight leaves patients at risk of receiving treatments that may not offer significant benefits and may even cause harm. Tightening regulation could help ensure that these therapies are only offered when there is clear scientific evidence supporting their use, thus protecting patients from being exposed to unnecessary risks. Moreover, increased regulation could help standardize practices across fertility clinics, ensuring that treatments are administered safely and consistently. By holding clinics and pharmaceutical companies accountable for the therapies they offer, tighter regulation could also help to reduce the commercialization of fertility treatments that exploit vulnerable patients for financial gain.

Additionally, regulatory oversight could improve access to these therapies for those who would otherwise be unable to afford them, by mandating insurance coverage or by offering subsidies for low-income patients. This would help reduce the disparity in access to care and ensure that fertility treatments are more equitably distributed among all individuals, regardless of socioeconomic status. The need for such regulatory oversight is especially urgent given the emotional and financial stakes involved in reproductive healthcare, where patients' decisions are often guided by hope and the desire for a successful pregnancy (Usadi & Merriam, 2015).

Overall, the commercialization of fertility and pregnancy-supporting immunotherapies presents significant ethical challenges, particularly regarding access to care and the regulation of unproven treatments. The high cost of these therapies, coupled with insurance implications, creates barriers to access that disproportionately affect lower-income individuals. The commercialization of these treatments raises concerns about the ethical integrity of fertility clinics and pharmaceutical companies,

who may prioritize profit over patient welfare. Tighter regulation of these therapies could help ensure that treatments are evidence-based, safe, and accessible to all patients, helping to mitigate the ethical risks associated with their commercialization. By improving oversight and standardizing practices, healthcare providers can better support patients in making informed decisions about their reproductive health.

***Ethical Guidelines and Policy Recommendations***

When clinicians offer experimental treatments such as intravenous immunoglobulin (IVIG) or intralipid therapy for recurrent pregnancy loss (RPL) or recurrent implantation failure (RIF), it is imperative that they provide patients with comprehensive, transparent, and understandable information. The ethical principle of informed consent requires that patients are fully aware of the risks, benefits, and uncertainties associated with these treatments before they make a decision (Beauchamp & Childress, 2019). Given the experimental nature of these therapies, clinicians must go beyond simply explaining the standard procedures and include specific information about the lack of rigorous clinical trials, the limited evidence for efficacy, and the potential for unknown side effects. Clinicians must also inform patients of alternative treatments or management strategies, whether they be conventional therapies or the option to forego treatment altogether. For example, a patient seeking treatment for RPL should be made aware of other non-immunological options, such as lifestyle changes or targeted hormone therapies, as well as the fact that many causes of infertility or pregnancy loss remain unexplained (Garmendia et al., 2025).

Additionally, clinicians should discuss financial costs and insurance implications, particularly since treatments like IVIG can be prohibitively expensive and are often not covered by insurance. This financial aspect can significantly impact a patient's autonomy, as the ability to afford treatment should not dictate a patient's decision-making capacity. The emotional vulnerabilities associated with infertility and pregnancy loss make it essential for clinicians to approach these discussions with sensitivity, ensuring that patients feel empowered to make decisions that align with their values. Clinicians should strive to create a non-coercive environment, where the patient feels supported in asking questions, expressing concerns, and exploring all options without feeling pressured to pursue any particular treatment due to hope or desperation (Lidz & Appelbaum, 2002).

Fertility clinics play a pivotal role in providing reproductive care, including experimental treatments like IVIG and intralipid therapy. However, the increasing commercialization of fertility services raises significant ethical concerns regarding the standard of care, the quality of information provided, and the potential exploitation of vulnerable patients. As reproductive medicine evolves, it is crucial that fertility clinics adhere to ethical oversight to ensure patient safety and equity. This oversight can be achieved through both regulatory mechanisms and internal ethical guidelines, ensuring that clinics maintain high standards for both clinical practice and patient care. Currently, many fertility clinics operate with minimal regulation, especially when offering unproven treatments. Some clinics may prioritize profit over patient well-being by offering experimental treatments that have not been conclusively proven to be effective. The ethical concern here lies in exploitation, where the emotional desperation of patients may be leveraged for financial gain. Patients may be offered therapies without sufficient evidence or informed consent, increasing the risk of harm and undermining the autonomy of patients (Usadi & Merriam, 2015). Therefore, it is essential to establish regulatory frameworks and external oversight to ensure that clinics provide only those treatments that are backed by sound scientific evidence or are undergoing rigorous clinical trials. Additionally, fertility clinics should implement ethical review boards or ethics committees that assess the risks and benefits of experimental treatments before they are offered to patients. These committees would ensure that any novel treatments are subjected to ethical scrutiny, considering not only the scientific validity but also the potential psychological and emotional impact on patients. Oversight can also help ensure that clinic staff adhere to ethical principles

such as justice (ensuring fair access to treatments) and non-maleficence (avoiding harm to patients) (Beauchamp & Childress, 2019).

Ethics committees play an essential role in evaluating and approving experimental treatments for reproductive health. The approval of such treatments should not solely rely on clinical efficacy but also on ethical considerations, including informed consent, patient autonomy, and the principle of justice. Ethical review committees should evaluate the evidence supporting the efficacy of treatments, as well as the potential risks to both the mother and the fetus, especially when the treatment in question is not yet fully validated by large-scale clinical trials (Røstvik, 2025). These committees must also ensure that the principle of autonomy is respected throughout the treatment process. This means ensuring that patients are fully informed of the experimental nature of treatments and are not coerced into accepting treatment options based on emotional distress or the manipulation of hope (Lidz & Appelbaum, 2002). Additionally, ethics committees should ensure that vulnerable populations, such as low-income women or those without adequate social support, are not disproportionately targeted or exploited by fertility clinics offering expensive or unproven therapies. They should ensure that patients have equal access to well-regulated and evidence-based care, avoiding scenarios where economic disparities determine treatment options.

To promote more equitable, informed, and autonomous decision-making in reproductive medicine, several policies should be implemented to change the outlook of reproductive care. Firstly, clear guidelines for informed consent: Clinicians must be required to adhere to standardized, comprehensive guidelines when obtaining informed consent, ensuring that all patients are fully aware of the experimental nature of the treatments, their potential risks, and the uncertainties involved. These guidelines should also require that the financial costs of treatment and insurance coverage options be fully disclosed to patients, ensuring that economic factors do not unduly influence their decisions (Usadi & Merriam, 2015). Secondly, stricter regulation of fertility clinics: Regulatory oversight of fertility clinics should be strengthened to prevent the promotion of unproven, experimental treatments. This could involve government oversight or independent accreditation bodies that evaluate the scientific rigor and ethical practices of clinics offering these treatments. Regulatory bodies should also ensure that clinics are transparent in their marketing and provide accurate information regarding the success rates and safety of experimental therapies (Røstvik, 2025). Thirdly, the establishment of ethics committees: Fertility clinics should establish internal ethics committees or collaborate with external ethics boards to review and approve any experimental treatments offered to patients. These committees would assess both the scientific evidence and the ethical implications of such treatments, ensuring that patients are not exposed to unnecessary risks and that their autonomy is respected throughout the treatment process.

Fourthly, public education and patient support: There should be a greater emphasis on patient education to ensure that individuals seeking fertility treatments have access to accurate and balanced information. Fertility clinics should provide patients with resources to help them make informed decisions, such as decision aids or counseling services. Public education campaigns could also be developed to inform the public about the potential risks and benefits of experimental fertility treatments, as well as alternative, evidence-based options. Fifthly, insurance coverage and financial assistance: Policies should be put in place to ensure that fertility treatments, including those used to support pregnancy and manage RPL and RIF, are covered under insurance plans. Financial assistance programs should also be available for individuals from low-income backgrounds, ensuring that access to these potentially life-altering treatments is not determined solely by financial status (Usadi & Merriam, 2015).
In reproductive medicine, especially when experimental treatments are involved, ethical guidelines and oversight are critical to ensuring that patient autonomy, equity, and informed decision-making are respected. Clinicians must provide clear, transparent information about treatment options and their associated risks, while fertility clinics must adhere to strong ethical standards and regulatory oversight

to prevent exploitation and ensure that all patients have equal access to safe, effective care. By implementing stricter regulations, enhancing informed consent processes, and ensuring support for vulnerable populations, the healthcare system can improve both the quality of care and the ethical integrity of reproductive medicine.

## CONCLUSION

This paper critically examined the ethical dilemmas surrounding the use of immunological interventions, such as intravenous immunoglobulin (IVIG) and intralipid therapy, in reproductive medicine. As we have seen, the use of these therapies to treat recurrent pregnancy loss (RPL) and recurrent implantation failure (RIF) offers hope to women who have faced multiple failures in traditional fertility treatments. However, this hope also comes with significant ethical concerns, particularly in the areas of informed consent, patient autonomy, and the commercialization of reproductive therapies.

The underlying issue lies in the complexity of maternal-fetal immune interactions, where disruptions in the delicate balance between immune tolerance and immune defense can lead to failed pregnancies. While therapies such as IVIG and intralipid have shown promise in some clinical settings, their lack of rigorous scientific validation raises concerns about the potential for therapeutic misconception, where patients may consent to these treatments based on incomplete or biased information. The pressure to pursue "every possible treatment" in the emotionally charged context of fertility challenges can cloud patients' judgment, compromising their ability to make informed decisions. This highlights the urgent need for fertility clinics and clinicians to provide clear, transparent, and evidence-based information to patients, ensuring that their decisions are based on a full understanding of the risks, benefits, and uncertainties of these therapies.

Moreover, the commercialization of fertility treatments introduces additional ethical complexities. The high cost of treatments like IVIG and intralipid, often not covered by insurance, creates barriers to access, disproportionately affecting lower-income individuals. The commercial interests of fertility clinics may lead to the promotion of experimental treatments that are not yet fully validated by large-scale trials, raising questions about the prioritization of patient welfare versus profit. These concerns underscore the need for stronger ethical oversight and regulation within the fertility industry, to prevent the exploitation of vulnerable patients and ensure that treatments offered are both safe and effective. This work also explored the tension between maternal and fetal interests, particularly when treatments necessary to support maternal health may harm the fetus. Bioethical principles such as autonomy, beneficence, non-maleficence, and justice must be carefully applied to navigate these complex decisions. Additionally, the framework of relational autonomy acknowledges that decision-making in reproductive contexts is often influenced by social and familial pressures, which must be taken into account to ensure truly informed and autonomous choices.

In light of these findings, several policy recommendations were put forth to promote more equitable, informed, and autonomous decision-making in reproductive medicine. These include the establishment of clear guidelines for informed consent, stronger regulation of fertility clinics, the creation of ethics committees to review experimental treatments, and increased public education and patient support. Furthermore, financial assistance programs and insurance coverage for fertility treatments would help to reduce the barriers created by cost, ensuring that these potentially life-altering therapies are accessible to all patients, regardless of their socioeconomic status. Ultimately, this thesis calls for a more balanced and ethically responsible approach to reproductive immunotherapy. By integrating scientific rigor with compassionate care, and by upholding the principles of justice, autonomy, and informed consent, we can ensure that reproductive healthcare advances in a way that respects the rights and dignity of all individuals, while promoting the best possible outcomes for both mothers and their children.

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